

Assignment 1

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$$1 \quad \frac{dx}{dt} = -2x + ax + \frac{x^3}{3}$$

Equilibrium Points

$$-2x + ax + \frac{x^3}{3} = 0$$

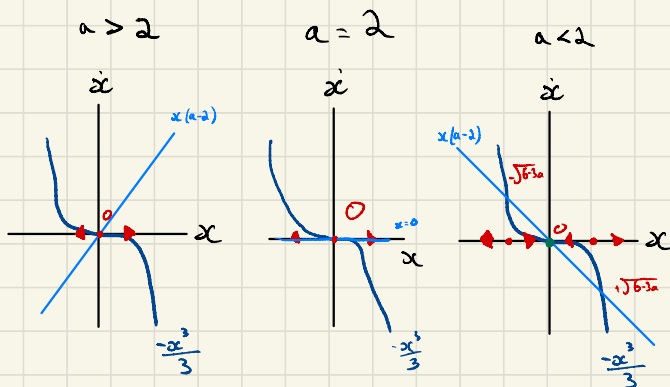
$$x(-2 + a + \frac{x^2}{3}) = 0$$

$$x^* = 0, \quad x = \pm \sqrt{3(a-2)}$$

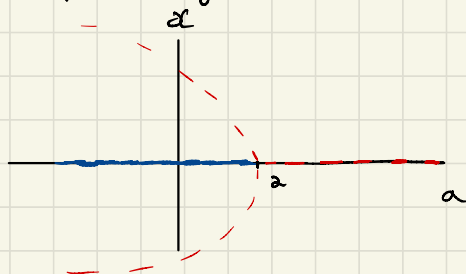
↳ only if $a \leq 0$
only if $a \leq 2$

$$x(a-2) > -\frac{x^3}{3}, \text{ moves right}$$

$$x(a-2) < -\frac{x^3}{3}, \text{ moves left}$$



Bifurcation diagram



Subcritical pitchfork bifurcation at point $(2, 0)$?

$$(b) \quad a = (-1) \cdot (8) + 0.5 = 8.5$$

$$\dot{x} = 6.5x + \frac{x^3}{3}$$

Equilibrium points:

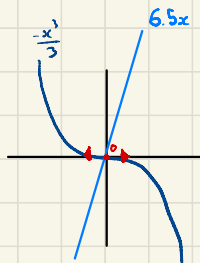
$$6.5x + \frac{x^3}{3} = 0$$

$$x^* = 0, \quad x^* = \pm \sqrt{-19.5}$$

Stability Analysis

$$6.5x > -\frac{x^3}{3} \quad \blacktriangleright$$

$$6.5x < -\frac{x^3}{3} \quad \blacktriangleleft$$



2.

$$\frac{dx}{dt} = 3x + 2y$$

$$\frac{dy}{dt} = -2x + 7y$$

Coefficient Matrice

$$\begin{pmatrix} 3 & 2 \\ -2 & 7 \end{pmatrix}$$

Eigenvalues

$$|A - \lambda I| = 0, \quad \begin{vmatrix} 3-\lambda & 2 \\ -2 & 7-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)(7-\lambda) + 4 = 0$$

$$\lambda^2 - 10\lambda + 25 = 0$$

$$(\lambda - 5)(\lambda - 5) = 0$$

$$\lambda = 5$$

Delta-Tau Plane

$$\Delta = 25$$

$$\tau = 10$$

$$\tau^2 - 4\Delta = 0$$

(check the Δ - τ diagram $\Rightarrow$ Ep(0,0) is an unstable degenerate node

Eigenvectors

$$v: A - \lambda I = \begin{pmatrix} -2 & 2 \\ -2 & 2 \end{pmatrix}, \quad v_1 = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$$

$$\text{Look for the unknown vector } \underline{p} \Rightarrow (A - \lambda I)\underline{p} = \underline{v} \Rightarrow \begin{pmatrix} -2 & 2 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \end{pmatrix} \Rightarrow -2p_x + 2p_y = -2$$

$$\underline{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

General Solution

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = c_1 e^{5t} \underline{v} + c_2 e^{5t} (\underline{v}t + \underline{p}) = c_1 e^{5t} \begin{pmatrix} -2 \\ -2 \end{pmatrix} + c_2 e^{5t} \left(t \begin{pmatrix} -2 \\ -2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$$

$$x(t) = -2c_1 e^{5t} + c_2 e^{5t}(-2t + 1) = (c_2 - 2c_1)e^{5t} - 2tc_2 e^{5t}$$

$$y(t) = -2c_1 e^{5t} + c_2 e^{5t}(-2t) = (-2c_1 - 2tc_2)e^{5t}$$

3. Phase Plain Portraits

Coefficient Matrix

$$A = \begin{pmatrix} -2 & 4 \\ -3 & 2 \end{pmatrix}$$

$$\frac{dx}{dt} = -2x + 4y$$

$$\frac{dy}{dt} = -3x + 2y$$

Eigenvalues

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -2-\lambda & 4 \\ -3 & 2-\lambda \end{vmatrix} = 0$$

$$(-2-\lambda)(2-\lambda) + 12 = 0$$

$$\lambda^2 + 8 = 0$$

$$\lambda = \pm \sqrt{-8} = \pm 2\sqrt{2}i$$

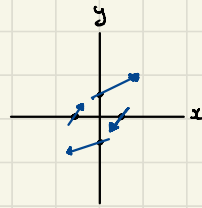
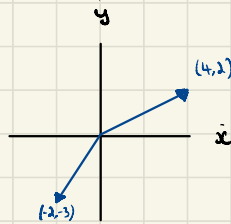
Delta-Tau plane

$$\Delta = 8 \quad (\text{check the } A-T \text{ diagram})$$

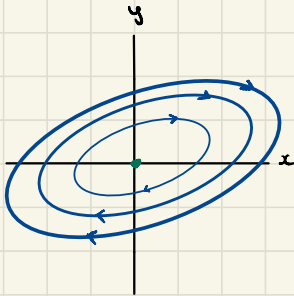
$\tau = 0$ Equilibrium point at $(0,0)$ is a Centre

Direction of Dynamics

(x, y)	$(\dot{x}, \dot{y})$
$(0, 1)$	$(4, 2)$
$(1, 0)$	$(-2, -3)$
$(0, -1)$	$(-4, -2)$
$(-1, 0)$	$(2, 3)$



Phase portrait



4. Parameter Dependence

(i)

$$\frac{dx}{dt} = (2-a)x + y$$

$$A = \begin{pmatrix} 2-a & 1 \\ -1 & 1 \end{pmatrix}$$

$$\frac{dy}{dt} = -x + y$$

Delta-Tau

$$\Delta = \det(A) = (2-a)(1) - (1)(-1) = 3-a$$

$$\tau = \text{tr}(A) = 2-a+1 = 3-a$$

$$\tau^2 - 4\Delta = (3-a)^2 - 4(3-a)$$

$$= 9 - 6a + a^2 - 12 + 4a = a^2 - 2a - 3$$

Critical Points

$$\Delta = 0 @ a = 3$$

$$\tau = 0 @ a = 3$$

$$\tau^2 - 4\Delta = a^2 - 2a - 3 = 0$$

$$(a+1)(a-3) = 0$$

$$a = -1, a = 3$$

Nature of Equilibrium point at (0,0) as a varies:

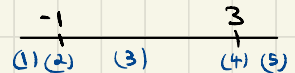
(1) $a < -1$, $\Delta > 0$, $\tau > 0$, $\tau^2 - 4\Delta > 0$: unstable node

(2) $a = -1$, $\Delta > 0$, $\tau > 0$, $\tau^2 - 4\Delta = 0$: degenerate unstable node

(3) $-1 < a < 3$, $\Delta > 0$, $\tau > 0$, $\tau^2 - 4\Delta < 0$: unstable spiral

(4) $a = 3$, $\Delta = 0$, $\tau = 0$, $\tau^2 - 4\Delta = 0$: non-isolated fixed points

(5) $a > 3$, $\Delta < 0$, $\tau < 0$, $\tau^2 - 4\Delta > 0$: saddle point



(ii) $a = (-1)^8 \cdot 8 + 0.5 = 8.5$

$$\frac{dx}{dt} = -6.5x + y$$

$$A = \begin{pmatrix} -6.5 & 1 \\ -1 & 1 \end{pmatrix}$$

$$\frac{dy}{dt} = -x + y$$

Delta-Tau

$$\Delta = \det(A) = -5.5$$

$$\tau = -5.5$$

$$\tau^2 - 4\Delta = 52.25$$

$\Delta < 0$, Equilibrium point at (0,0) is a saddle point